

```
public static int task6CM(int[] input) {
    int n = input.length;
    int maxLength = 0;
    int[] dp = new int[n];
    for (int i = 0; i < n; i++) {
        dp[i] = 1;
        for (int j = 0; j < i; j++) {
            if (input[i] > input[j]) {
                dp[i] = Math.max(dp[i], dp[j] + 1);
            }
        }
    }
    for (int length : dp) {
        maxLength = Math.max(maxLength, length);
    }
    return maxLength;
}

public static void main(String[] args) {
    int[] input = {10, 9, 2, 5, 3, 7, 101, 18};
    int result = task6CM(input);
    System.out.println(result);
}
```